

ARES 5

For metal parts, ARES reports axial stress (σ_x) and shear stress (τ_{xy}) for each element and section.

The static margin is obtained as Reserve Factor comparing the material ultimate stress (F_{tu}) with the Von Mises stress (σ_{vm}), where this is defined, for the quasi-unidimensional ARES problem ($\sigma_y=0$), as:

$$\sigma_{vm} = (\sigma_x^2 + 3 \cdot \tau_{xy}^2)^{0.5}$$

If analysis requires the principal stresses, the following expressions can be used to do the transformation:

$$\sigma_1 = \sigma_x/2 + [(\sigma_x/2)^2 + \tau_{xy}^2]^{0.5}$$

$$\sigma_2 = \sigma_x/2 - [(\sigma_x/2)^2 + \tau_{xy}^2]^{0.5}$$

ARES 4

In the previous version, ARES reported the Von Mises stress (one data column instead of 2 for stresses), but also the resultant section load (P), combining axial load (N) and shear load (S). Then, taking into account section width (w) and thickness (t), as...

$$\sigma_{vm}^2 = \sigma_x^2 + 3 \cdot \tau_{xy}^2$$

$$P^2 = N^2 + S^2 \rightarrow [P/(w \cdot t)]^2 = [N/(w \cdot t)]^2 + [S/(w \cdot t)]^2 = \sigma_x^2 + \tau_{xy}^2$$

A simple linear system of two equations can be solved to get the axial and shear stresses:

$$\sigma_{vm}^2 = \sigma_x^2 + 3 \cdot \tau_{xy}^2$$

$$[P/(w \cdot t)]^2 = \sigma_x^2 + \tau_{xy}^2$$

...

$$\sigma_x^2 = \{ 3 \cdot [P/(w \cdot t)]^2 - \sigma_{vm}^2 \} / 2$$

$$\tau_{xy}^2 = \{ \sigma_{vm}^2 - [P/(w \cdot t)]^2 \} / 2$$

The transformation to principal stresses can be done as shown in previous section.